

SPRINT-04. Travel Planner Remote Persistence

Abstract

In this sprint, students will focus on **integrating Retrofit** to connect the app with a REST API for hotel reservations, hosted at <http://15.224.84.148:8090>. Students will implement screens to consume endpoints of the demo API created by the professor.

The goal is to implement **hotel search, booking, reservation listing and cancellation**, and also **image gallery management** for each trip.

Students are also expected to:

- Implement hotel search for **London, Paris or Barcelona** using **start and end dates**, consuming the **availability** endpoint.
- Display a list of hotels and rooms (typically 3), allowing the user to **book a room**.
- **Save the reservation info locally** in the database as a **new trip**.
- Show **hotel and room images** in the search and booking screens.
- List all reservations and allow **deleting them**.
- Create an **image gallery** feature per trip, saving the images **locally**.

Objectives

- Understand and apply Retrofit for HTTP calls.
- Connect with external APIs using android.
- Manage images and gallery

Tasks

T1. Retrofit Configuration (3 Points)

- T1.1: Add Retrofit dependency and configure the HTTP client.
- T1.2: Create the necessary data models and API interfaces to access the hotel API implementing MVVM structure.
- T1.3: Create a repository layer to abstract the API usage.
- T1.4: Create unit tests mocking the remote connection.

T2. Search and Booking Screens (5 Points)

- T2.1: Create a screen to search for hotels in **London, Paris, or Barcelona**, with city, start and end dates.
- T2.2: Show hotel and room data returned by the API (typically 3 rooms per hotel).
- T2.3: Allow users to **book a room** and save the reservation info (ID, room, hotel, price, etc) locally in the trip table or another table creating a new trip and save all reservation info.
- T2.4: Display **all images** of the hotel and rooms on the booking.

Observation: follow the structure used in the example seen in class which implements:

- Search screen with a list of available hotels/rooms
- Hotel/room details screen with a button to book a room

T3. Add Images / Gallery to Trip (4 Points)

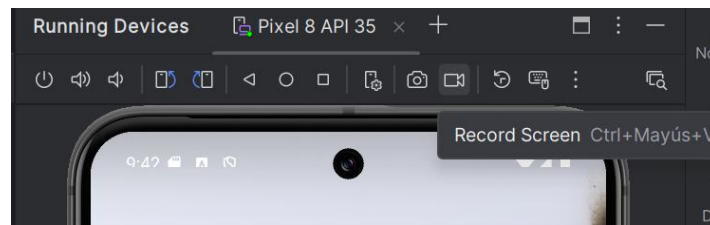
- T3.1: Allow the user to attach multiple images to a trip.
- T3.2: Save images locally in the device database or storage.
- T3.3: Display trip-specific galleries in the trip details screen.

T4. List and Cancel Reservations (3 Points)

- T4.1: Create a screen to **list all local reservations** indicating the trip related.
- T4.2: Add functionality to **delete a reservation** locally and via API (if required).
- T4.3: Show the associated hotel and room images in the list, similar to the example seen in class (RemotePersistence.zip)
- T4.4: Update the "My Trips" screen to indicate whether a trip includes a hotel reservation, and display the corresponding reservation details.

Delivery

- Students must submit the GitHub link containing a repository with all tasks completed and the respective release/tag. The Sprint.md document must list task assignments for each team member, ensuring everyone contributes.
- Ensure this release is the v4.x.x. Remember, only the latest version will be tested.
- Record a video demonstrating your application running showing all implemented tasks using the Android Studio emulator recording feature or your mobile phone recording feature if you are using it to test. Save the video in the following directory: /docs or documentation/evidence/v4.x.x
- MUST USE Repository structure and HILT as DI library



Attention:

The delivery will be penalized by 20% of the overall grade per item if any of the following conditions are not met:

- Dependency Injection is not implemented using HILT or Koin.
- Persistence is not implemented using ROOM.
- The standard architecture (View – ViewModel – Repository – DB) is not followed, unless an alternative architecture has been explicitly approved by the professor.
- The folder structure does not include at least the following: view, viewmodel, repo, di, and data.
- All date and time fields must be implemented using date pickers.
- The project name is still incorrect in Android Studio.

Sprint 04. Applications for mobile devices**Course:** 105025-2526**Institution:** University of Lleida (Campus Igualada)**Professor:** Vítor da Silva Verbel

Evaluation Criteria

Metric	Unsatisfactory (factor: 0.4)	Acceptable (0.6)	Very Good (1)	Excellent (1.1)
Clarity	Code lacks documentation and is unreadable.	Some documentation, but style is poor.	Well-documented, but minor readability issues.	Fully documented, well-structured, and easy to read.
Modularity	Poorly structured functions with little logic.	Divided into appropriate units but lacks cohesion.	Mostly well-structured with minimal redundancy.	Highly modular, reusable, and efficient code.
Completeness	Major functionality missing or poorly handled.	Some case handling but lacks robustness.	Mostly complete, minor cases may be missing.	Fully functional with all cases handled properly.
Testing	No tests or minimal test coverage.	Tests covering partially a major functionality.	Tests covering all major functionalities.	Tests covering all major and minor functionalities.